



Mobile Communications by Schiller

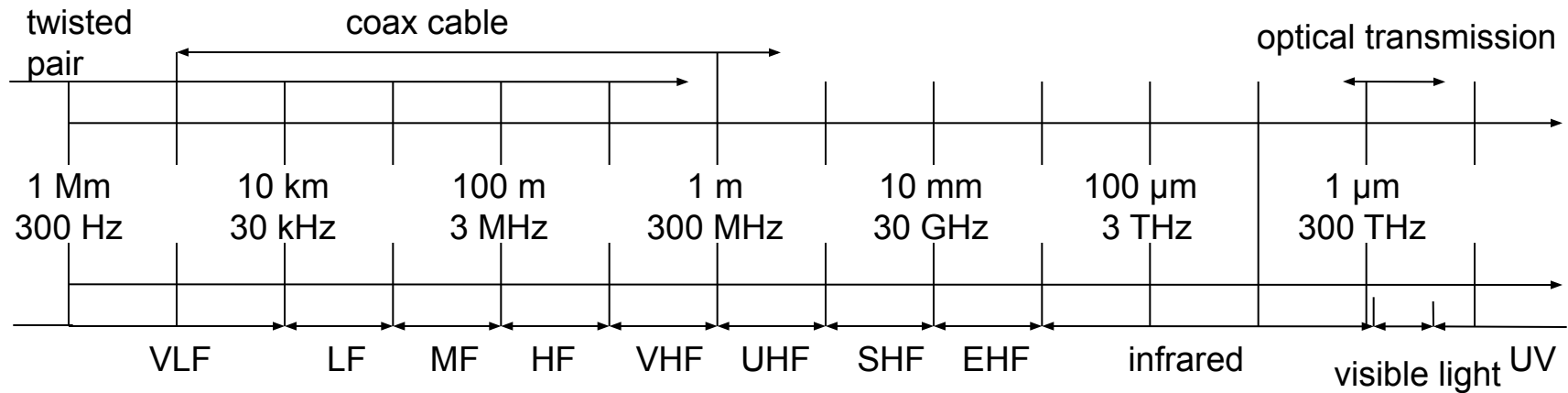
Chapter 2: Wireless Transmission

- ☐ Frequencies
- ☐ Signals
- ☐ Antenna
- ☐ Signal propagation
- ☐ Multiplexing
- ☐ Spread spectrum
- ☐ Modulation
- ☐ Cellular systems





Frequencies for communication



VLF = Very Low Frequency

LF = Low Frequency

MF = Medium Frequency

HF = High Frequency

VHF = Very High Frequency

UHF = Ultra High Frequency

SHF = Super High Frequency

EHF = Extra High Frequency

UV = Ultraviolet Light

Frequency and wave length:

$$\lambda = c/f$$

wave length λ , speed of light $c \approx 3 \times 10^8 \text{ m/s}$, frequency f





Frequencies for mobile communication

- ❑ VHF-/UHF-ranges for mobile radio
 - ❑ simple, small antenna for cars
 - ❑ deterministic propagation characteristics, reliable connections
- ❑ SHF and higher for directed radio links, satellite communication
 - ❑ small antenna, focussing
 - ❑ large bandwidth available
- ❑ Wireless LANs use frequencies in UHF to SHF spectrum
 - ❑ some systems planned up to EHF
 - ❑ limitations due to absorption by water and oxygen molecules (resonance frequencies)
 - weather dependent fading, signal loss caused by heavy rainfall etc.





Frequencies and regulations

ITU-R holds auctions for new frequencies, manages frequency bands worldwide (WRC, World Radio Conferences)

	Europe	USA	Japan
Mobile phones	NMT 453-457MHz, 463-467 MHz; GSM 890-915 MHz, 935-960 MHz; 1710-1785 MHz, 1805-1880 MHz	AMPS, TDMA, CDMA 824-849 MHz, 869-894 MHz; TDMA, CDMA, GSM 1850-1910 MHz, 1930-1990 MHz;	PDC 810-826 MHz, 940-956 MHz; 1429-1465 MHz, 1477-1513 MHz
Cordless telephones	CT1+ 885-887 MHz, 930-932 MHz; CT2 864-868 MHz DECT 1880-1900 MHz	PACS 1850-1910 MHz, 1930-1990 MHz PACS-UB 1910-1930 MHz	PHS 1895-1918 MHz JCT 254-380 MHz
Wireless LANs	IEEE 802.11 2400-2483 MHz HIPERLAN 1 5176-5270 MHz	IEEE 802.11 2400-2483 MHz	IEEE 802.11 2471-2497 MHz





Signals I

- ❑ physical representation of data
- ❑ function of time and location
- ❑ signal parameters: parameters representing the value of data
- ❑ classification
 - ❑ continuous time/discrete time
 - ❑ continuous values/discrete values
 - ❑ analog signal = continuous time and continuous values
 - ❑ digital signal = discrete time and discrete values
- ❑ signal parameters of periodic signals:
period T , frequency $f=1/T$, amplitude A , phase shift ϕ
 - ❑ sine wave as special periodic signal for a carrier:

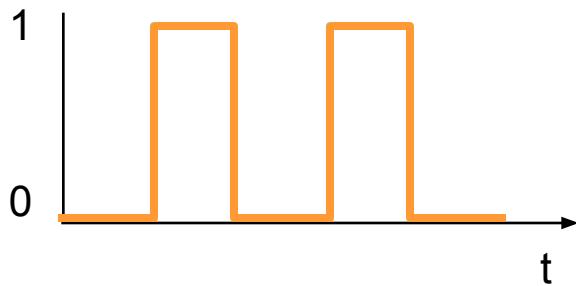
$$s(t) = A_t \sin(2 \pi f_t t + \phi_t)$$



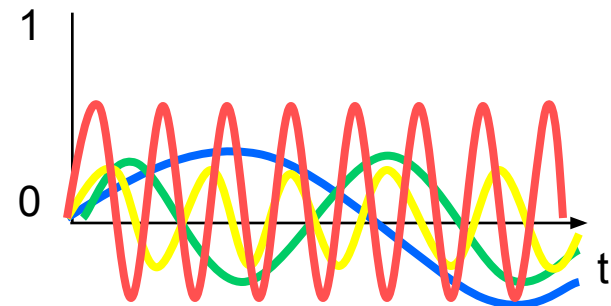


Fourier representation of periodic signals

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$



ideal periodic signal



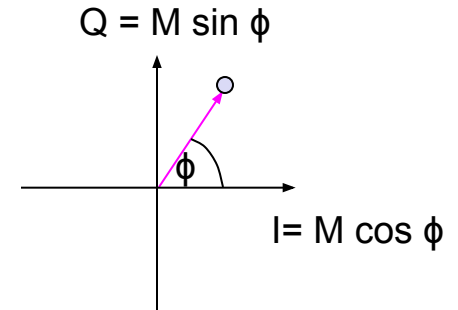
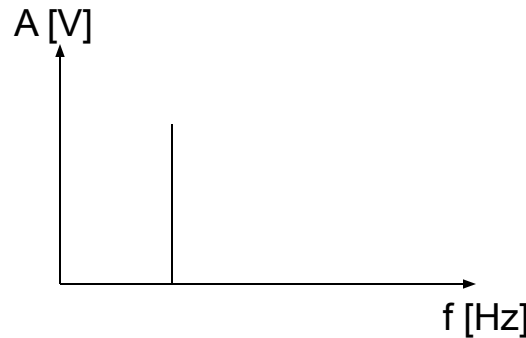
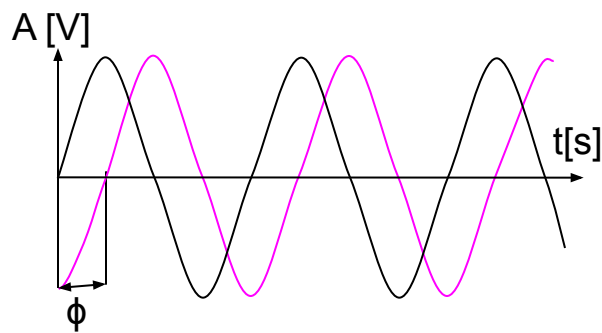
real composition
(based on harmonics)





Signals II

- ❑ Different representations of signals
 - ❑ amplitude (amplitude domain)
 - ❑ frequency spectrum (frequency domain)
 - ❑ phase state diagram (amplitude M and phase ϕ in polar coordinates)



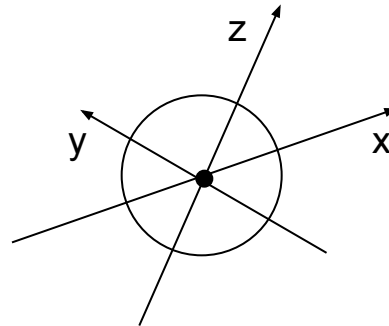
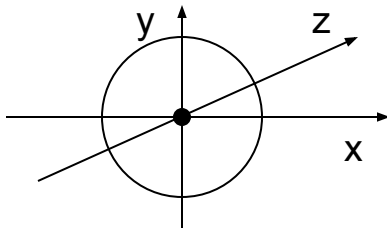
- ❑ Composed signals transferred into frequency domain using Fourier transformation
- ❑ Digital signals need
 - ❑ infinite frequencies for perfect transmission
 - ❑ modulation with a carrier frequency for transmission (analog signal!)





Antennas: isotropic radiator

- ❑ Radiation and reception of electromagnetic waves, coupling of wires to space for radio transmission
- ❑ Isotropic radiator: equal radiation in all directions (three dimensional) - only a theoretical reference antenna
- ❑ Real antennas always have directive effects (vertically and/or horizontally)
- ❑ Radiation pattern: measurement of radiation around an antenna



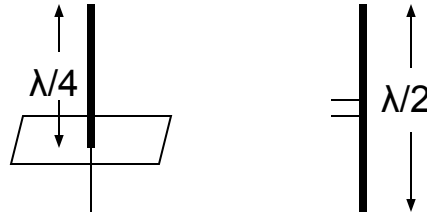
ideal
isotropic
radiator



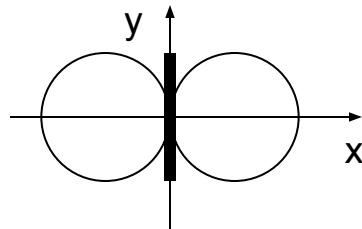


Antennas: simple dipoles

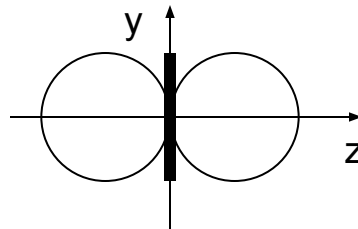
- ❑ Real antennas are not isotropic radiators but, e.g., dipoles with lengths $\lambda/4$ on car roofs or $\lambda/2$ as Hertzian dipole
 - ❑ shape of antenna proportional to wavelength



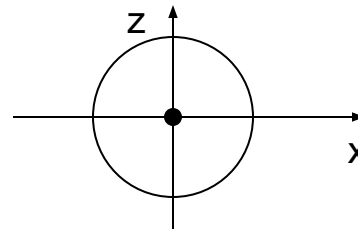
- ❑ Example: Radiation pattern of a simple Hertzian dipole



side view (xy-plane)



side view (yz-plane)



top view (xz-plane)

simple
dipole

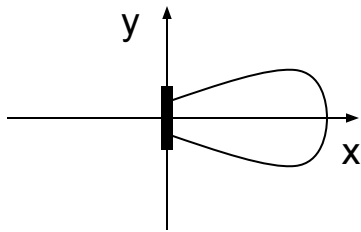
- ❑ Gain: maximum power in the direction of the main lobe compared to the power of an isotropic radiator (with the same average power)



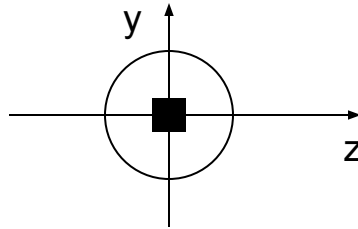


Antennas: directed and sectorized

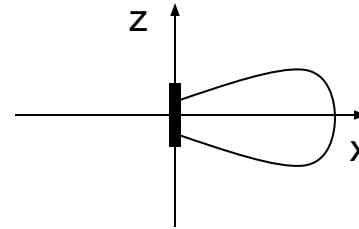
Often used for microwave connections or base stations for mobile phones (e.g., radio coverage of a valley)



side view (xy-plane)

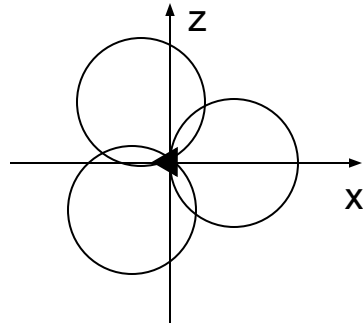


side view (yz-plane)

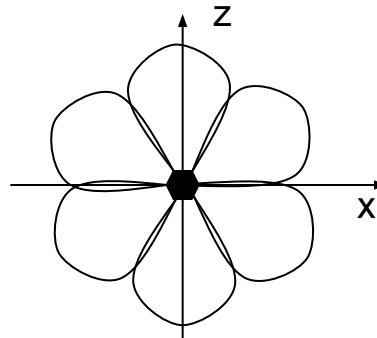


top view (xz-plane)

directed
antenna



top view, 3 sector



top view, 6 sector

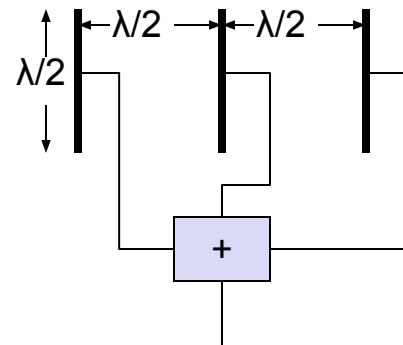
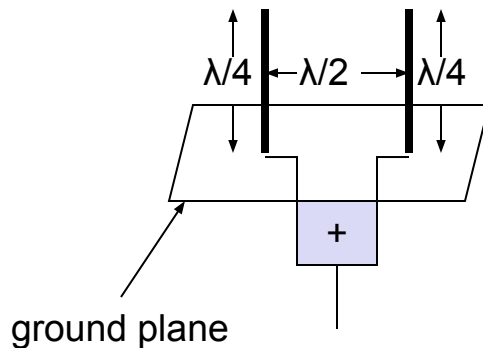
sectorized
antenna





Antennas: diversity

- ❑ Grouping of 2 or more antennas
 - ❑ multi-element antenna arrays
- ❑ Antenna diversity
 - ❑ switched diversity, selection diversity
 - receiver chooses antenna with largest output
 - ❑ diversity combining
 - combine output power to produce gain
 - cophasing needed to avoid cancellation





Types of Antennas:

1. Wire antennas
2. Aperture antennas
3. Reflector antennas
4. Lens antennas
5. Microstrip antennas
6. Array antennas

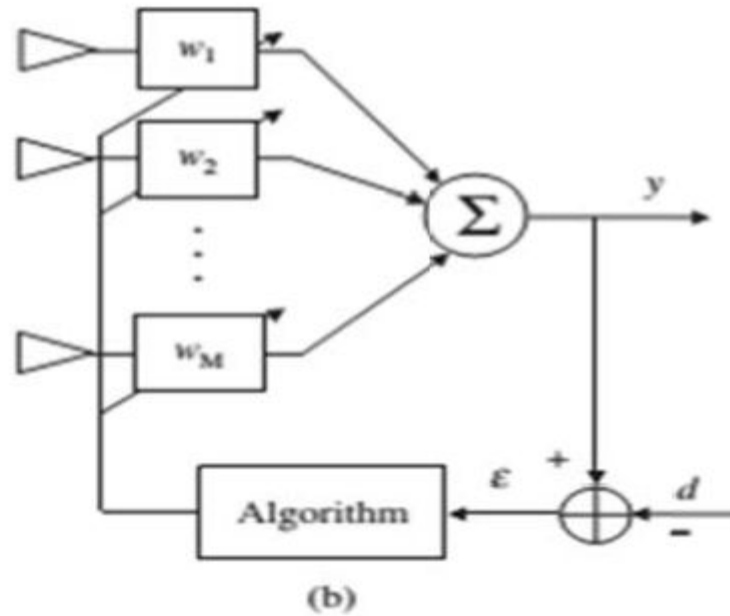
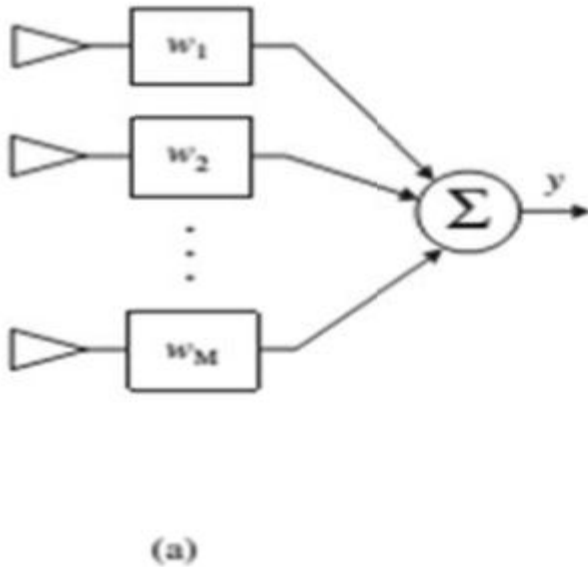




Smart antennas

Smart antenna system combines

1. Antenna array technology with
2. DSP algorithms to make the antenna system smart





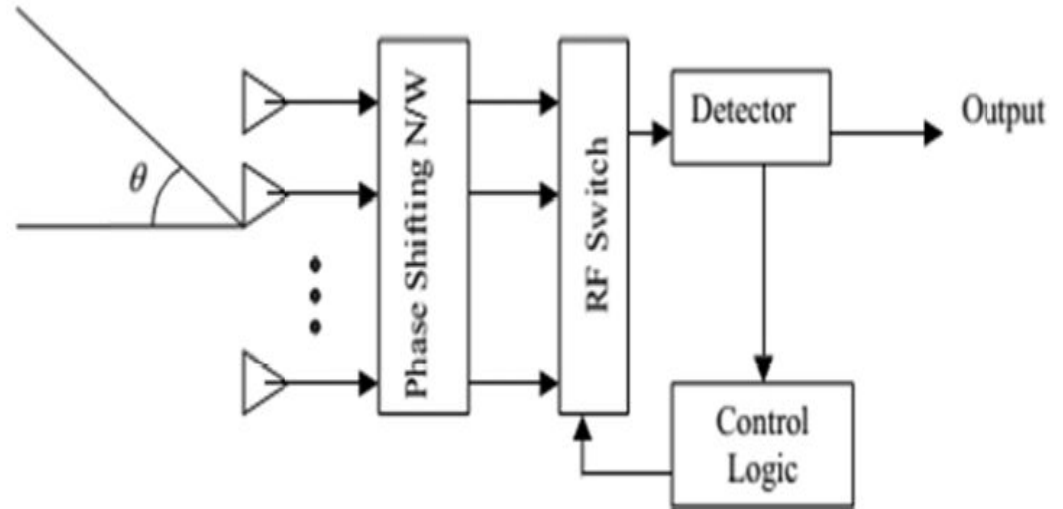
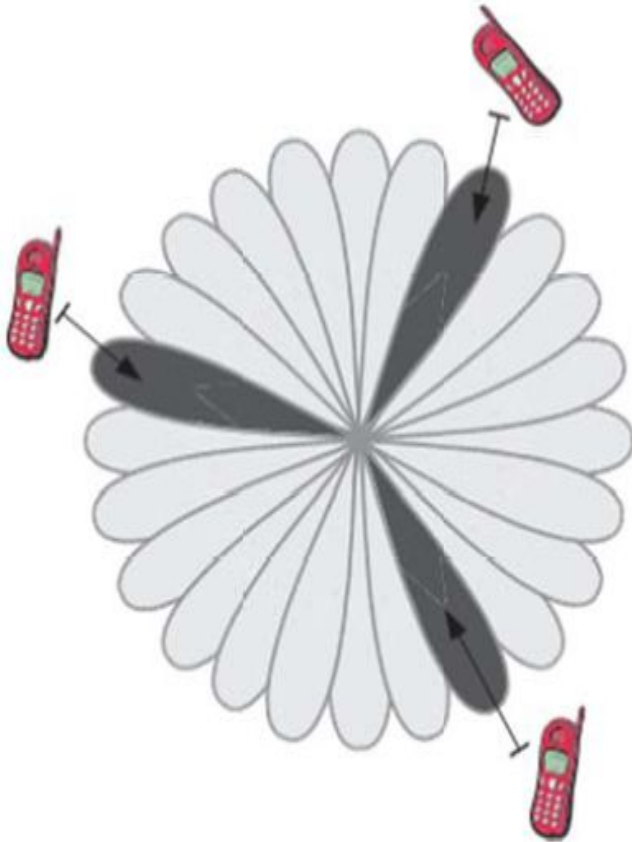
Types of Smart Antenna Systems

1. Switched-Beam: A finite number of fixed, predefined patterns
2. Adaptive Array: A theoretically infinite number of patterns (scenario-based) that are adjusted in real time according to the spatial changes of SOIs and SNOIs.



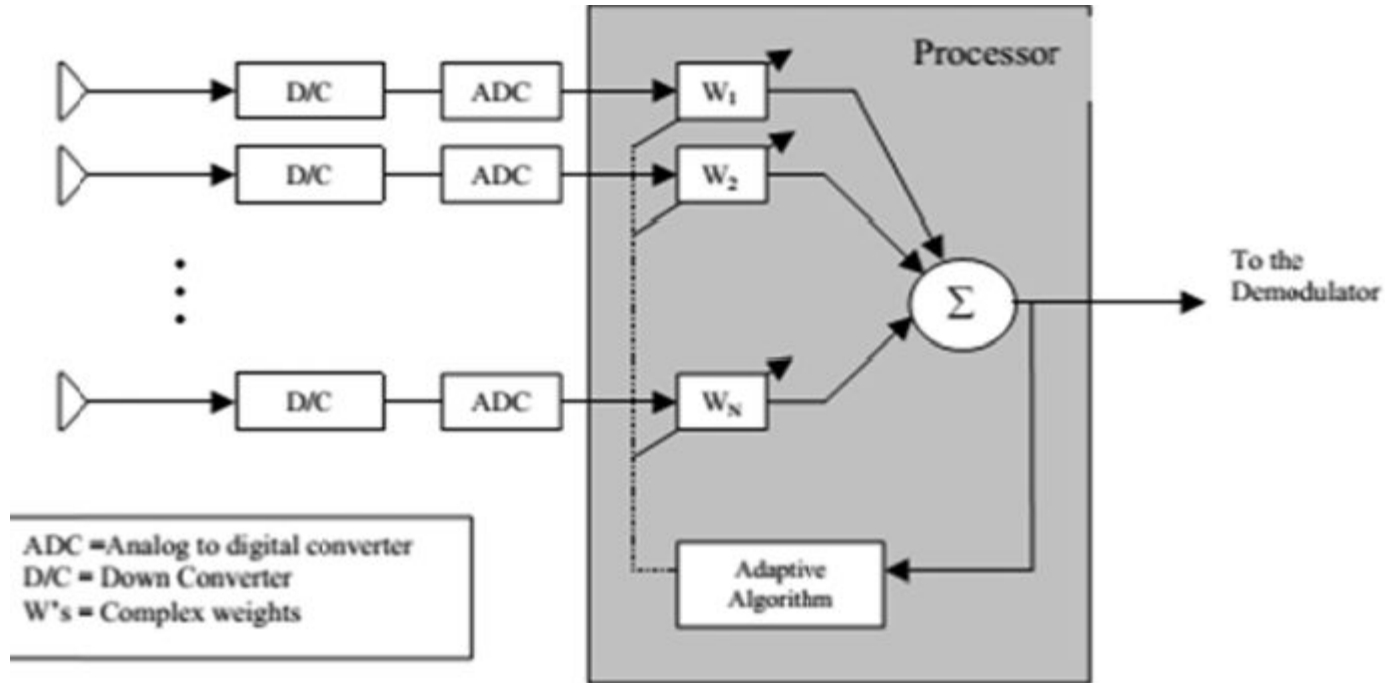


Switched-Beam





Adaptive Array





Signal propagation ranges

Transmission range

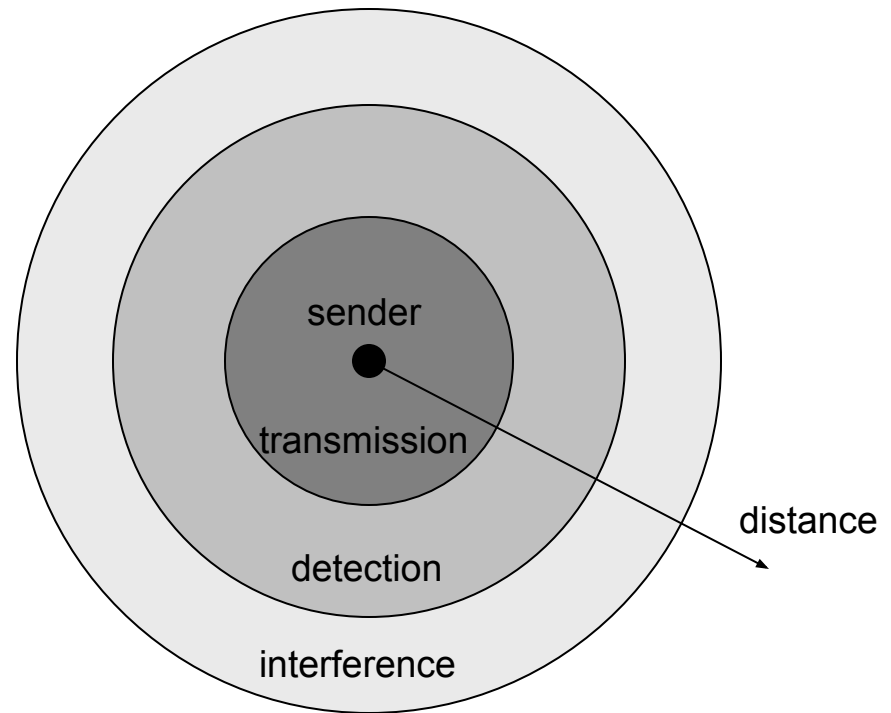
- ☐ communication possible
- ☐ low error rate

Detection range

- ☐ detection of the signal possible
- ☐ no communication possible

Interference range

- ☐ signal may not be detected
- ☐ signal adds to the background noise





Signal propagation

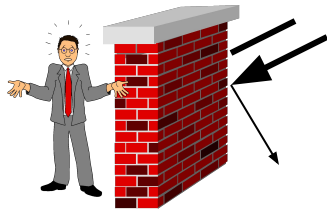
Propagation in free space always like light (straight line)

Receiving power proportional to $1/d^2$

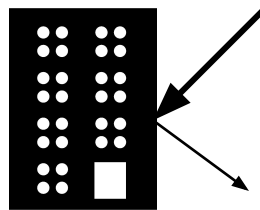
(d = distance between sender and receiver)

Receiving power additionally influenced by

- ☐ fading (frequency dependent)
- ☐ shadowing
- ☐ reflection at large obstacles
- ☐ scattering at small obstacles
- ☐ diffraction at edges



shadowing



reflection



scattering



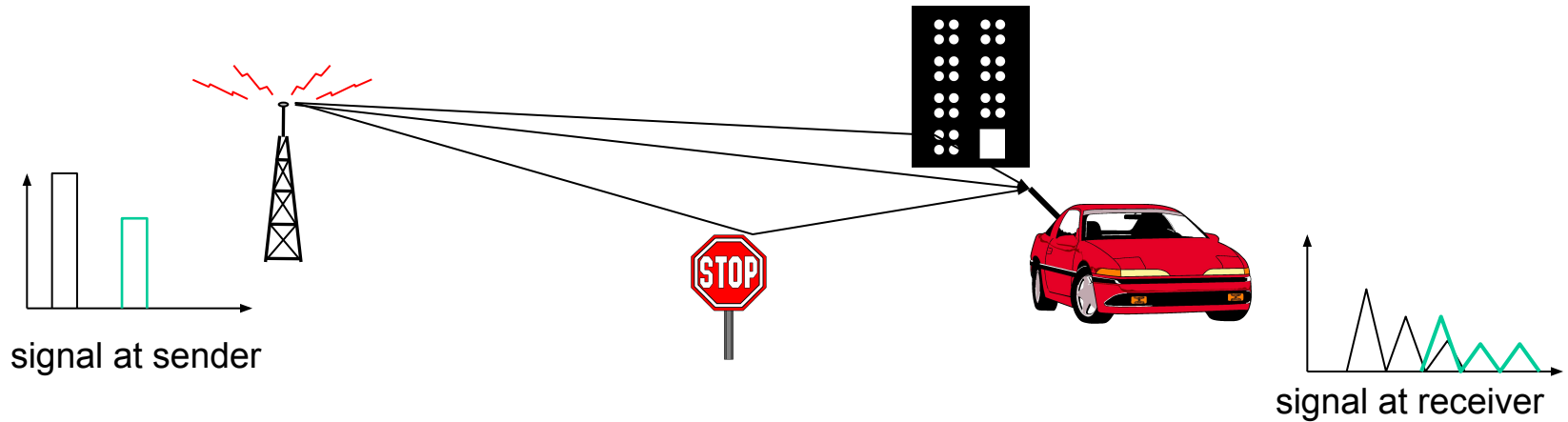
diffraction





Multipath propagation

Signal can take many different paths between sender and receiver due to reflection, scattering, diffraction



Time dispersion: signal is dispersed over time

- interference with “neighbor” symbols, Inter Symbol Interference (ISI)

The signal reaches a receiver directly and phase shifted

- distorted signal depending on the phases of the different parts





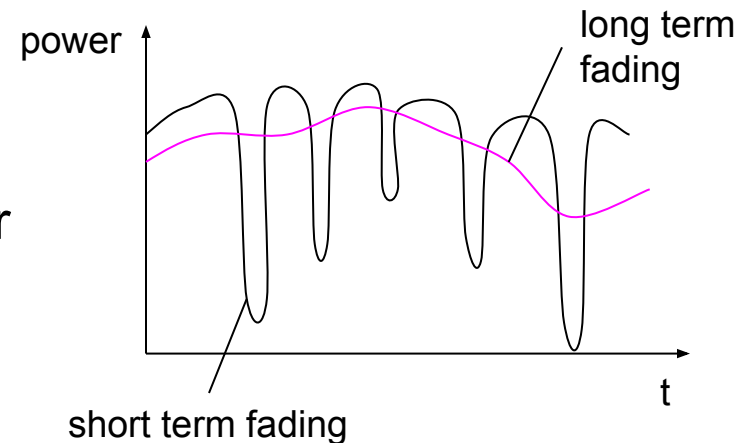
Effects of mobility

Channel characteristics change over time and location

- ❑ signal paths change
- ❑ different delay variations of different signal parts
- ❑ different phases of signal parts
- ❑ quick changes in the power received (short term fading)

Additional changes in

- ❑ distance to sender
- ❑ obstacles further away
- ❑ slow changes in the average power received (long term fading)





Multiplexing

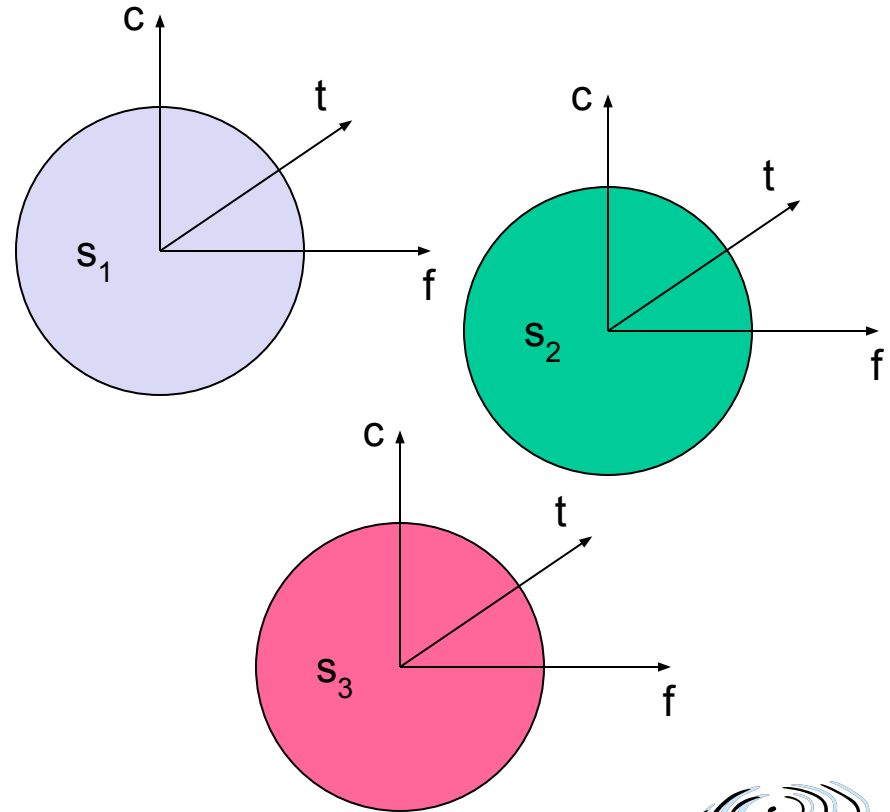
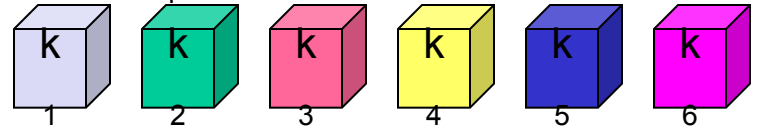
Multiplexing in 4 dimensions

- ❑ space (s_i)
- ❑ time (t)
- ❑ frequency (f)
- ❑ code (c)

Goal: multiple use
of a shared medium

Important: guard spaces needed!

channels k_i





Frequency multiplex

Separation of the whole spectrum into smaller frequency bands

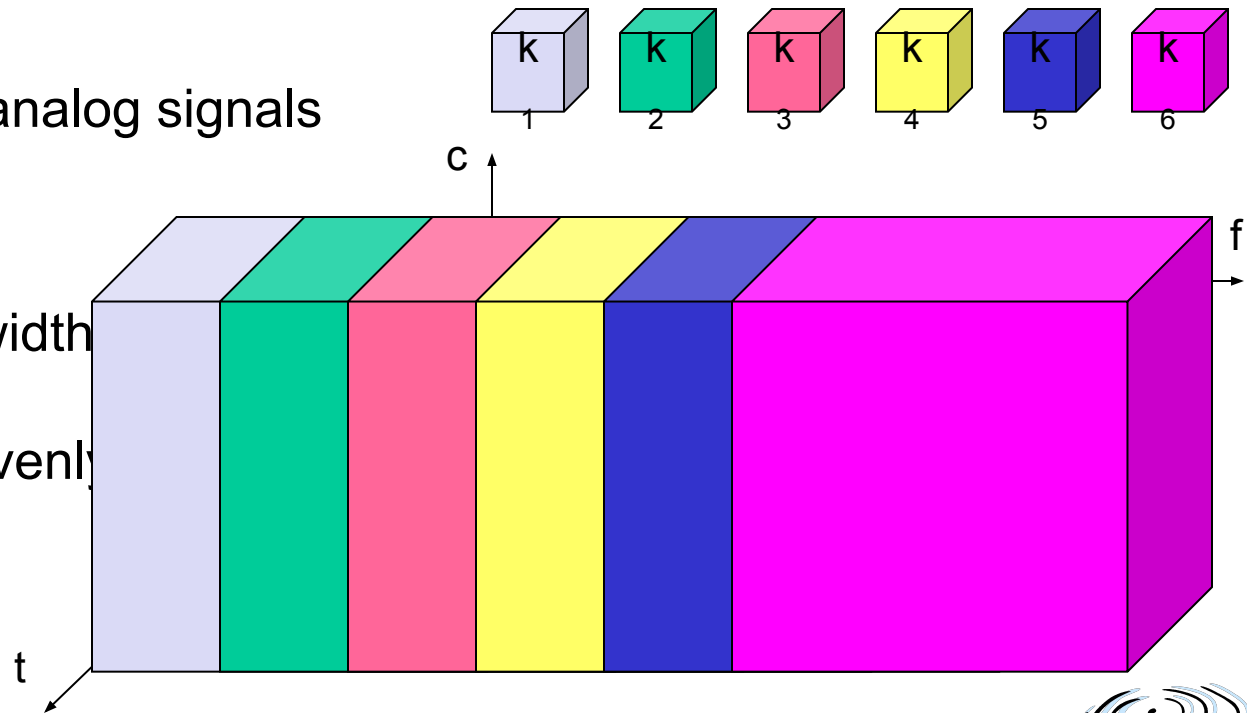
A channel gets a certain band of the spectrum for the whole time

Advantages:

- ❑ no dynamic coordination necessary
- ❑ works also for analog signals

Disadvantages:

- ❑ waste of bandwidth if the traffic is distributed unevenly
- ❑ inflexible
- ❑ guard spaces



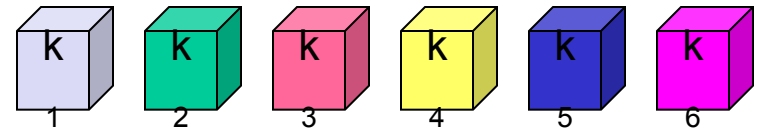


Time multiplex

A channel gets the whole spectrum for a certain amount of time

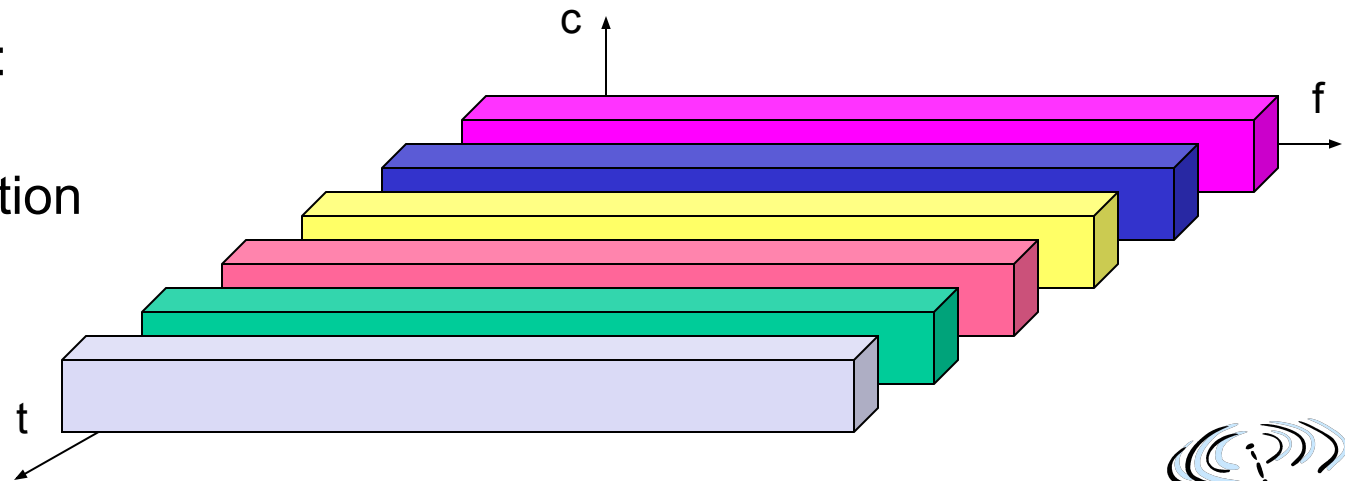
Advantages:

- ❑ only one carrier in the medium at any time
- ❑ throughput high even for many users



Disadvantages:

- ❑ precise synchronization necessary





Time and frequency multiplex

Combination of both methods

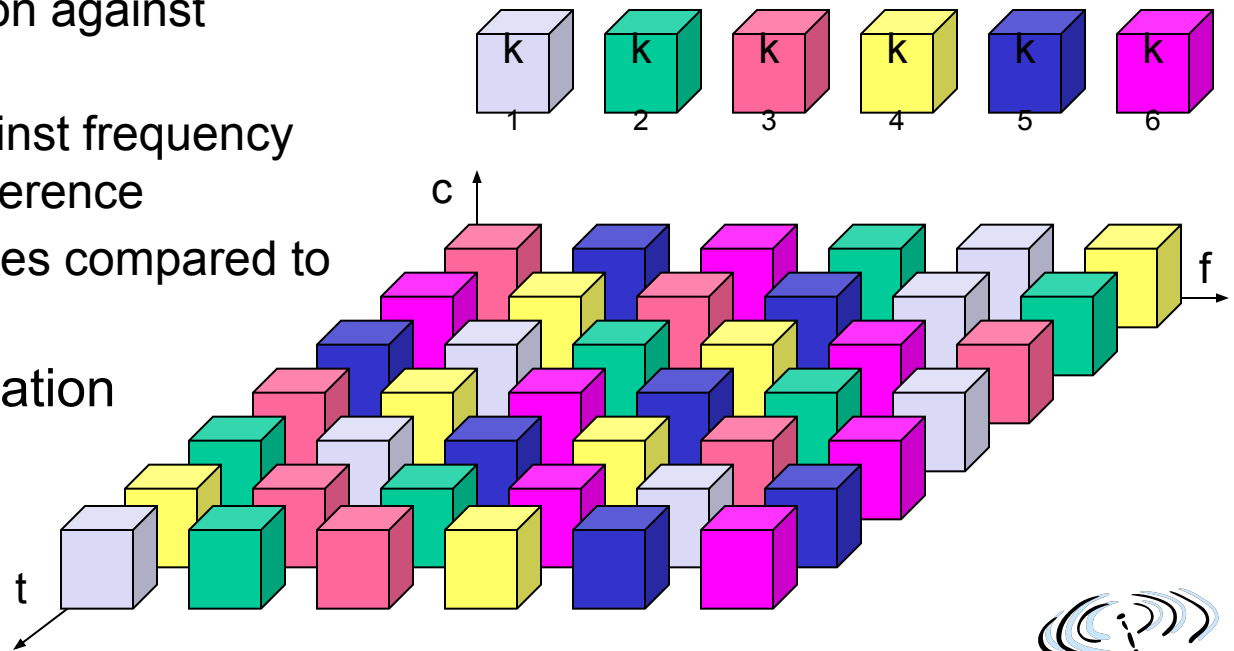
A channel gets a certain frequency band for a certain amount of time

Example: GSM

Advantages:

- ❑ better protection against tapping
- ❑ protection against frequency selective interference
- ❑ higher data rates compared to code multiplex

but: precise coordination required





Code multiplex

Each channel has a unique code

All channels use the same spectrum at the same time

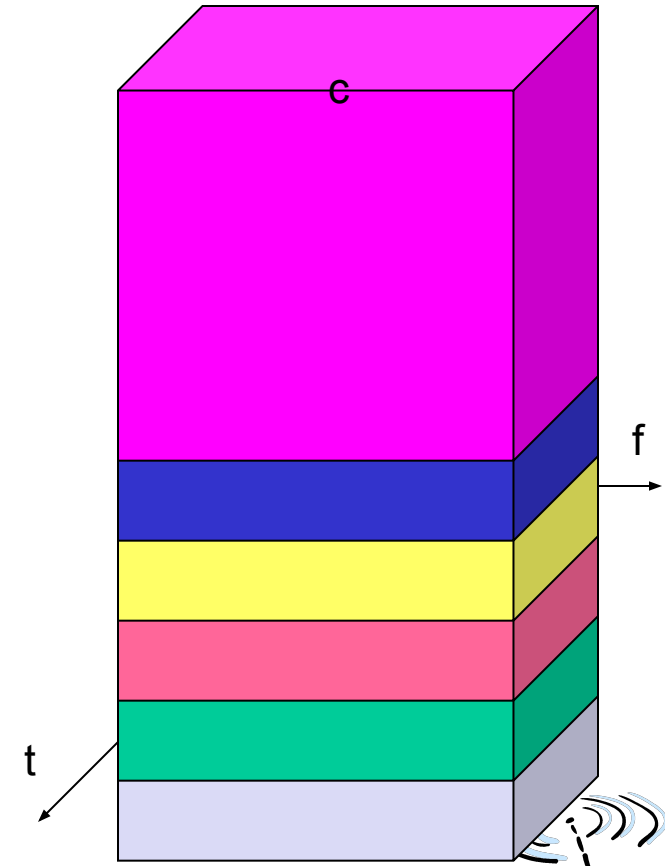
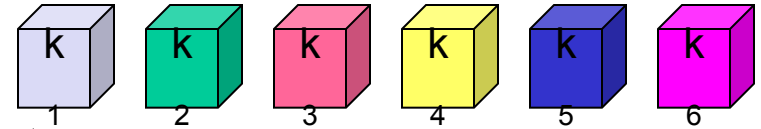
Advantages:

- ☐ bandwidth efficient
- ☐ no coordination and synchronization necessary
- ☐ good protection against interference and tapping

Disadvantages:

- ☐ lower user data rates
- ☐ more complex signal regeneration

Implemented using spread spectrum technology





Modulation

Digital modulation

- ❑ digital data is translated into an analog signal (baseband)
- ❑ ASK, FSK, PSK - main focus in this chapter
- ❑ differences in spectral efficiency, power efficiency, robustness

Analog modulation

- ❑ shifts center frequency of baseband signal up to the radio carrier

Motivation

- ❑ smaller antennas (e.g., $\lambda/4$)
- ❑ Frequency Division Multiplexing
- ❑ medium characteristics

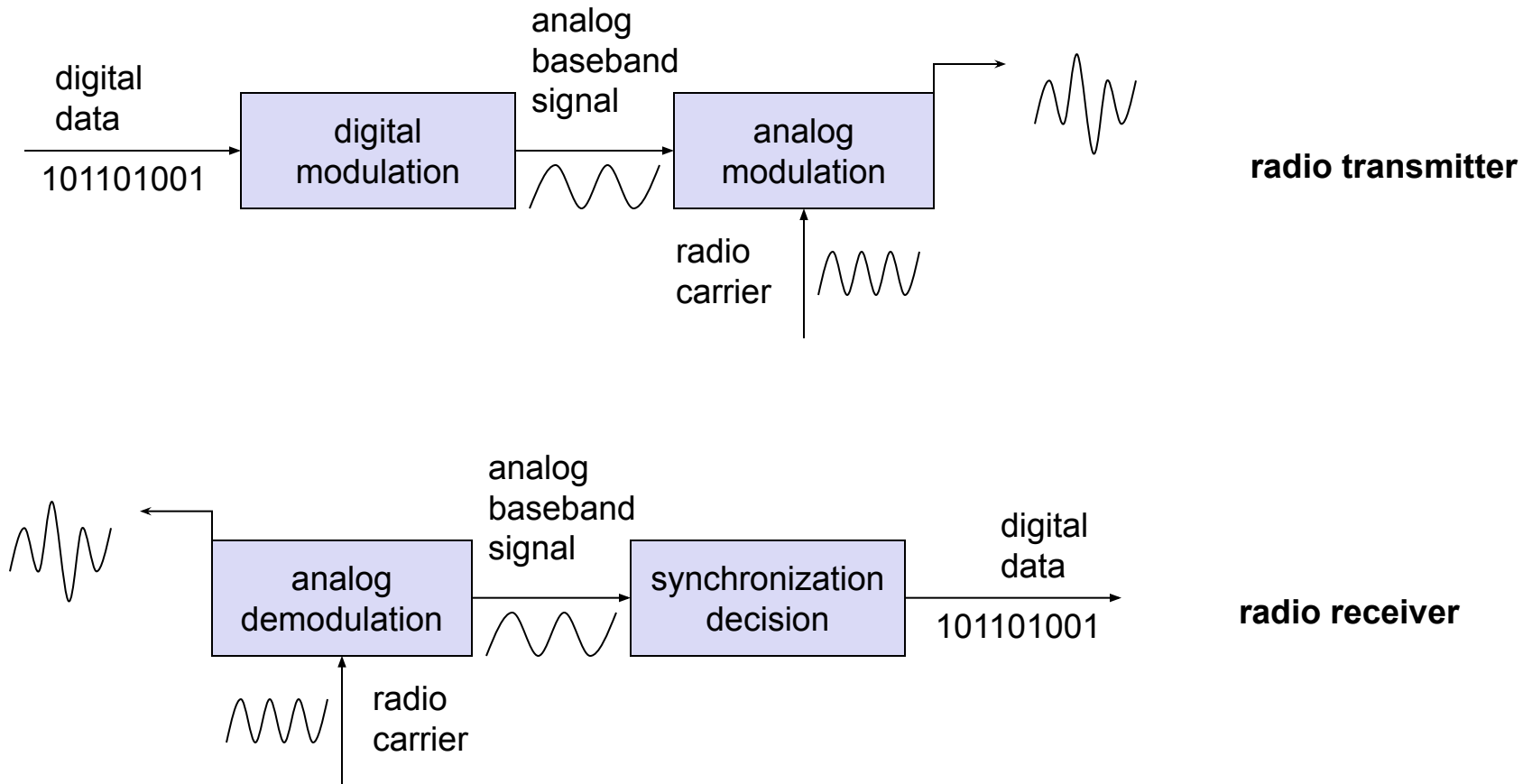
Basic schemes

- ❑ Amplitude Modulation (AM)
- ❑ Frequency Modulation (FM)
- ❑ Phase Modulation (PM)





Modulation and demodulation

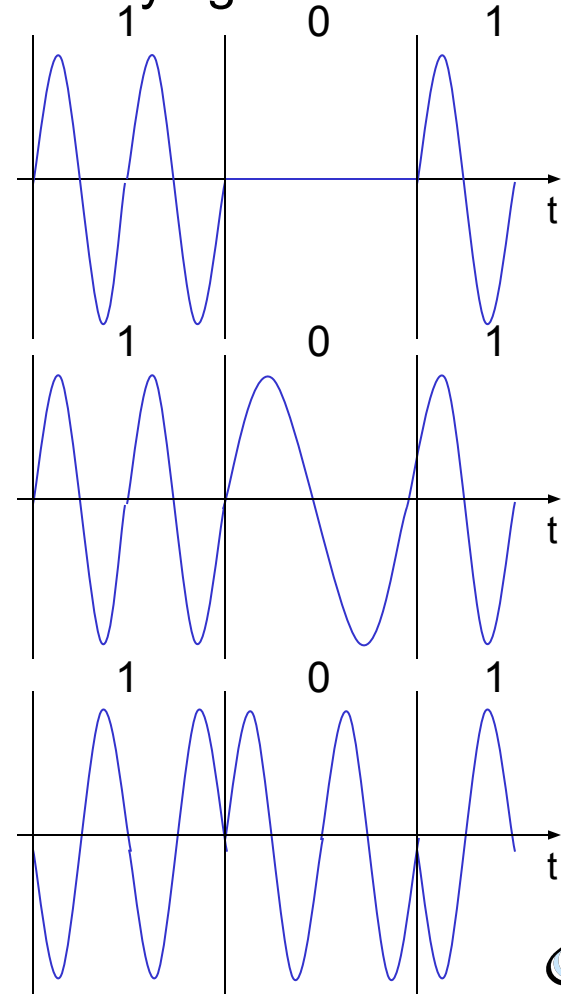




Digital modulation

Modulation of digital signals known as Shift Keying

- ❑ Amplitude Shift Keying (ASK):
 - ❑ very simple
 - ❑ low bandwidth requirements
 - ❑ very susceptible to interference
- ❑ Frequency Shift Keying (FSK):
 - ❑ needs larger bandwidth
- ❑ Phase Shift Keying (PSK):
 - ❑ more complex
 - ❑ robust against interference





Advanced Frequency Shift Keying (MSK)

In a first step, data bits are separated into even and odd bits, the duration of each bit being doubled.

The scheme also uses two frequencies: f_1 , the lower frequency, and f_2 , the higher frequency, with $f_2 = 2f_1$.





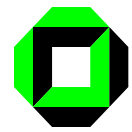
Rules of MSK

According to the following scheme, the lower or higher frequency is chosen (either inverted or non-inverted) to generate the MSK signal:

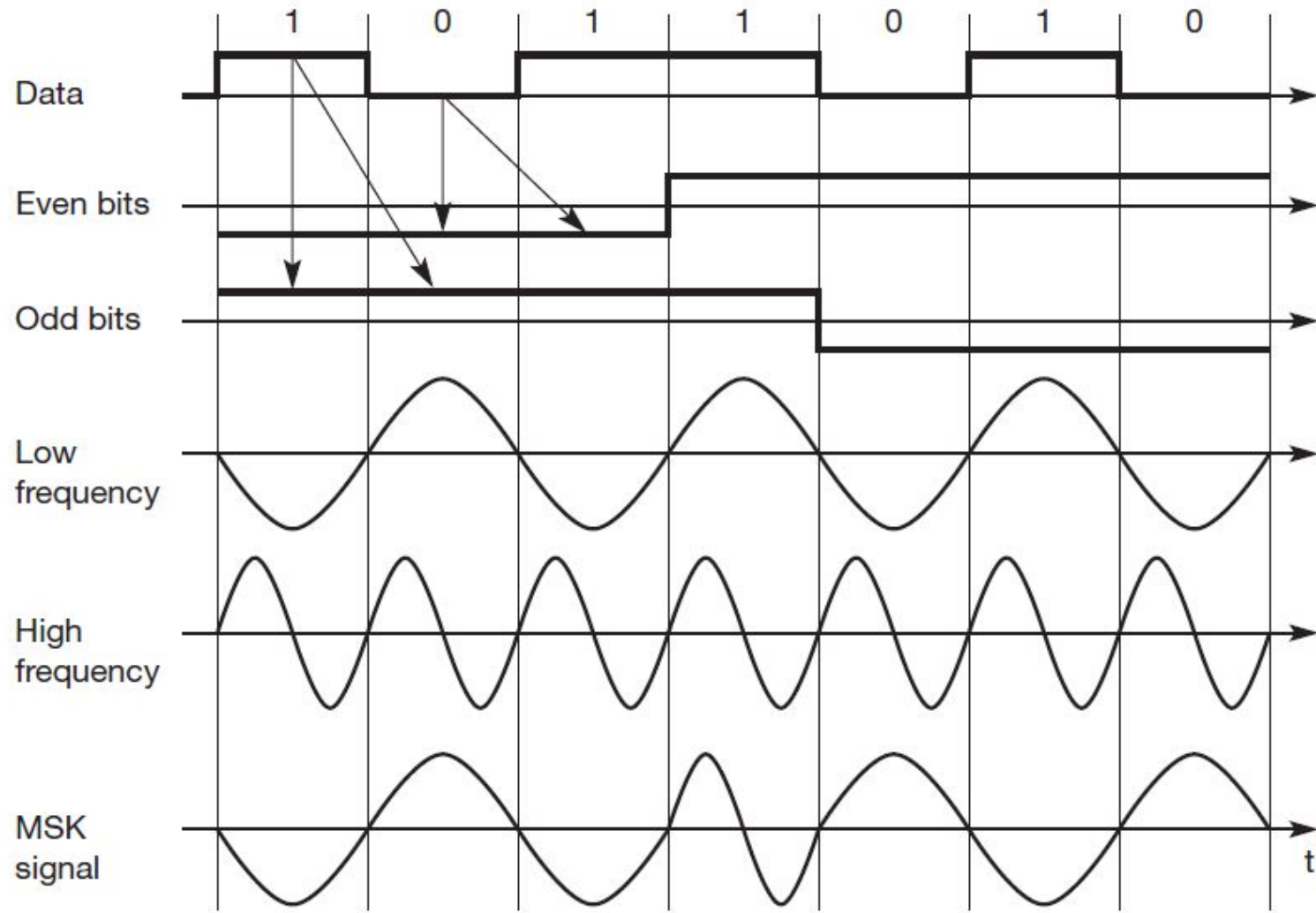
- if the even and the odd bit are both 0, then the higher frequency f_2 is inverted (i.e., f_2 is used with a phase shift of 180°);
- if the even bit is 1, the odd bit 0, then the lower frequency f_1 is inverted. This is the case, e.g., in the fifth to seventh columns of Figure (Example of MSK),
- if the even bit is 0 and the odd bit is 1, as in columns 1 to 3, f_1 is taken without changing the phase,
- if both bits are 1 then the original f_2 is taken.

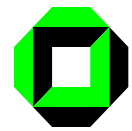
A high frequency is always chosen if even and odd bits are equal.
The signal is inverted if the odd bit equals 0. This scheme avoids all phase shifts in the resulting MSK signal.



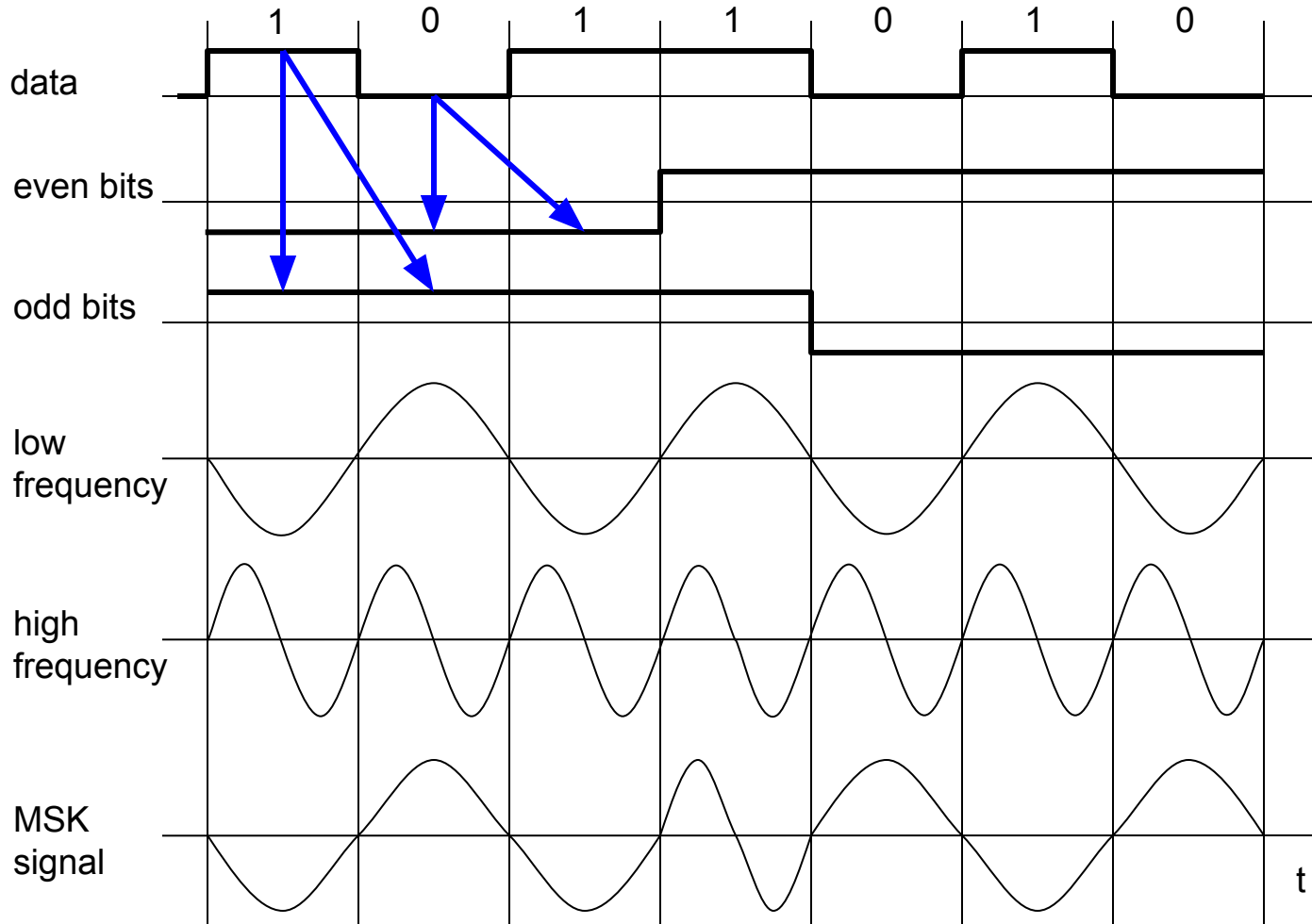


Example of MSK





Example of MSK



bit	
even	0 1 0 1
odd	0 0 1 1
signal value	- - + +
	h n n h

h: high frequency
n: low frequency
+: original signal
-: inverted signal

No phase shifts!





Advanced Phase Shift Keying

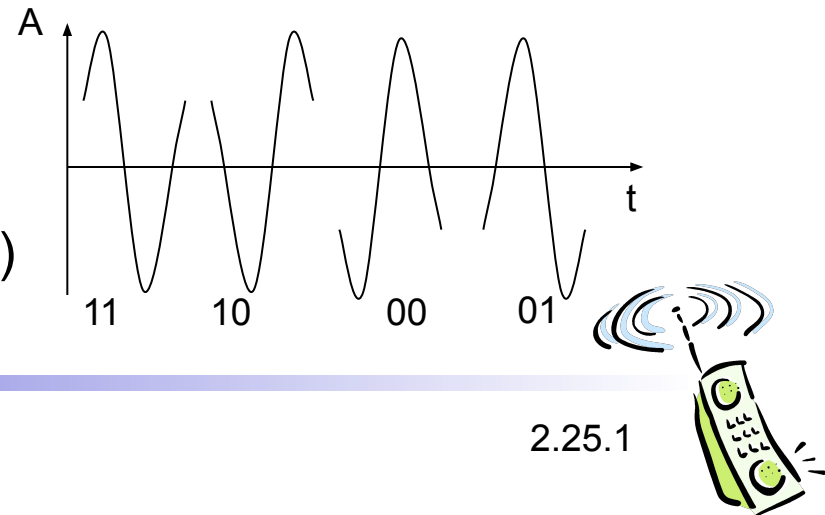
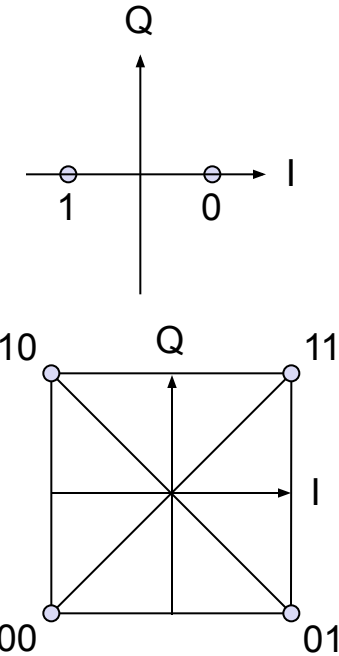
BPSK (Binary Phase Shift Keying):

- ❑ bit value 0: sine wave
- ❑ bit value 1: inverted sine wave
- ❑ very simple PSK
- ❑ low spectral efficiency
- ❑ robust, used e.g. in satellite systems

QPSK (Quadrature Phase Shift Keying):

- ❑ 2 bits coded as one symbol
- ❑ symbol determines shift of sine wave
- ❑ needs less bandwidth compared to BPSK
- ❑ more complex

Often also transmission of relative, not absolute phase shift: DQPSK - Differential QPSK (IS-136, PACS, PHS)

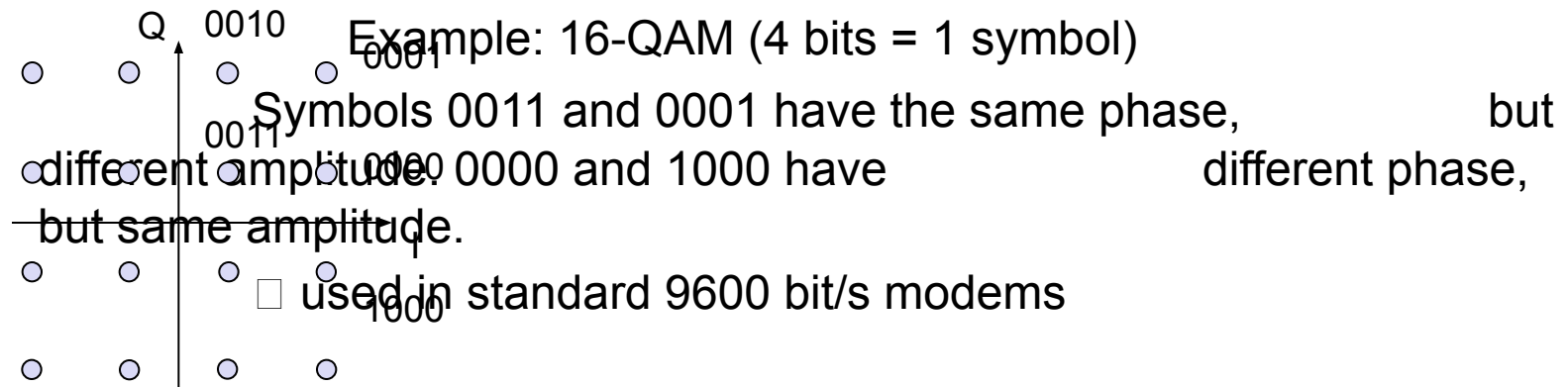




Quadrature Amplitude Modulation

Quadrature Amplitude Modulation (QAM): combines amplitude and phase modulation

- ❑ it is possible to code n bits using one symbol
- ❑ 2^n discrete levels, $n=2$ identical to QPSK
- ❑ bit error rate increases with n , but less errors compared to comparable PSK schemes



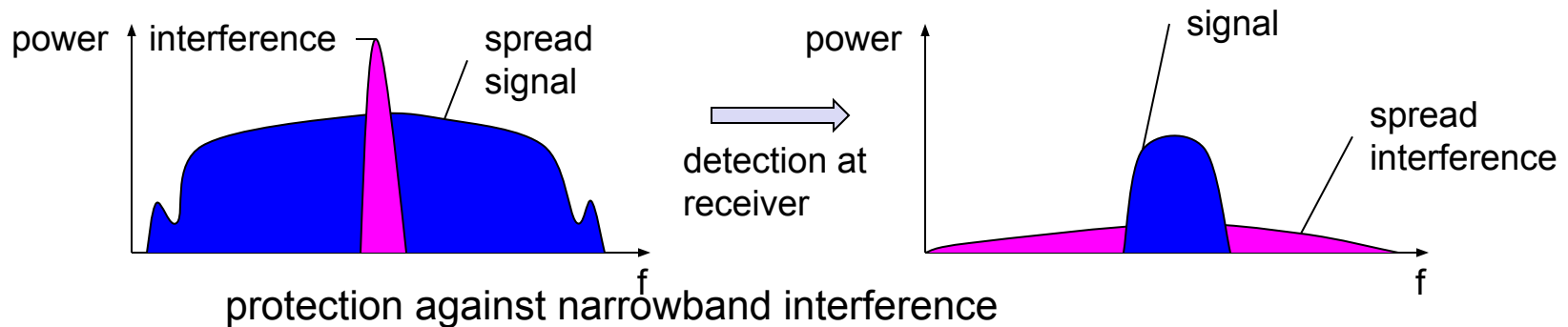


Spread spectrum technology

Problem of radio transmission: frequency dependent fading can wipe out narrow band signals for duration of the interference

Solution: spread the narrow band signal into a broad band signal using a special code

protection against narrow band interference



Side effects:

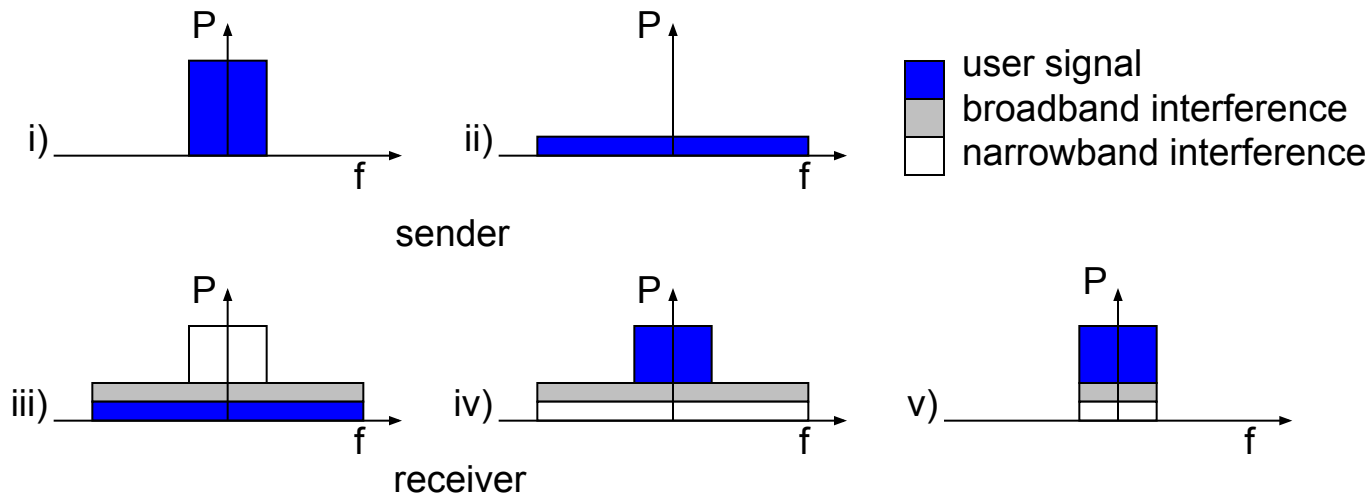
- ❑ coexistence of several signals without dynamic coordination
- ❑ tap-proof

Alternatives: Direct Sequence, Frequency Hopping



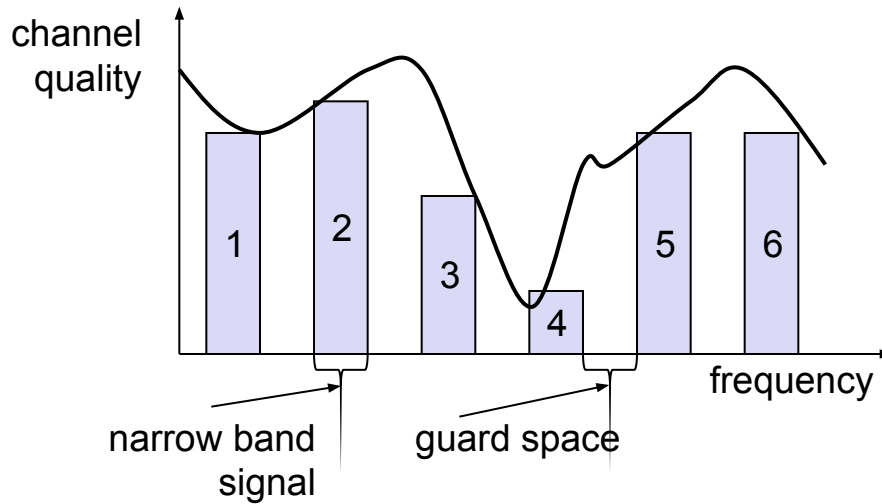


Effects of spreading and interference

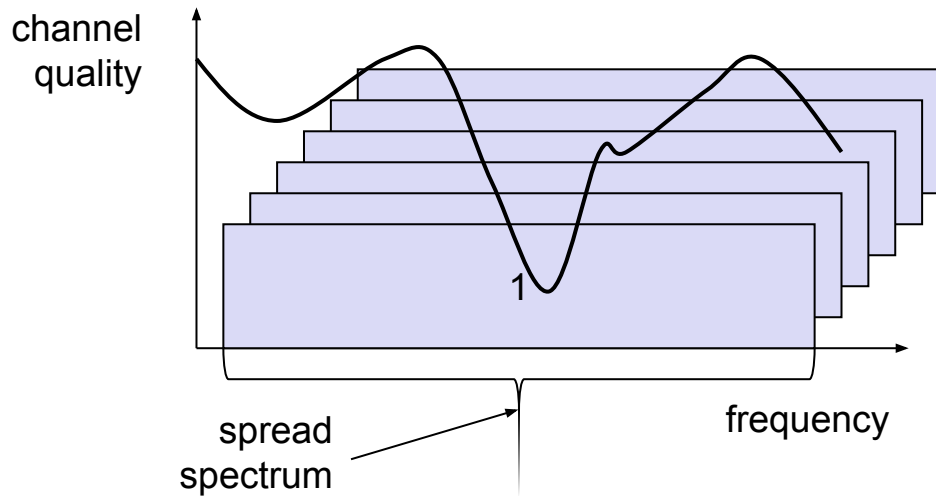




Spreading and frequency selective fading



narrowband channels



spread spectrum channels





DSSS (Direct Sequence Spread Spectrum) I

XOR of the signal with pseudo-random number (chipping sequence)

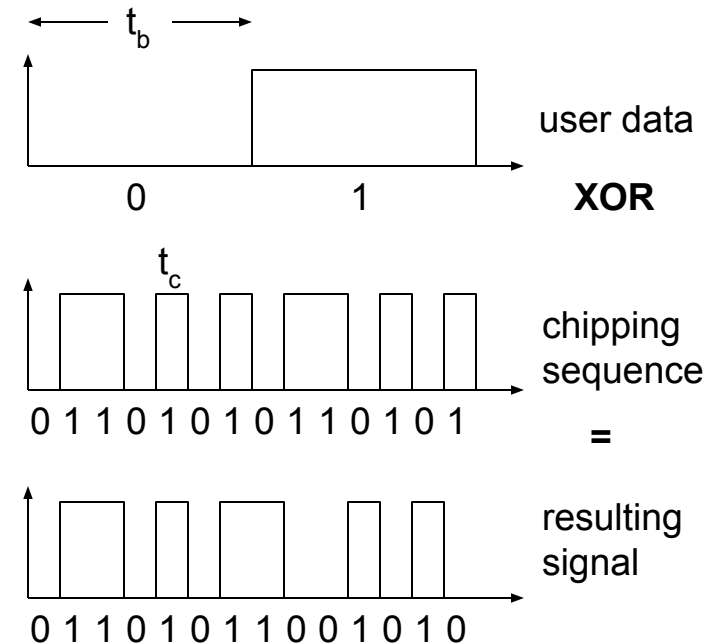
- ❑ many chips per bit (e.g., 128) result in higher bandwidth of the signal

Advantages

- ❑ reduces frequency selective fading
- ❑ in cellular networks
 - base stations can use the same frequency range
 - several base stations can detect and recover the signal
 - soft handover

Disadvantages

- ❑ precise power control necessary

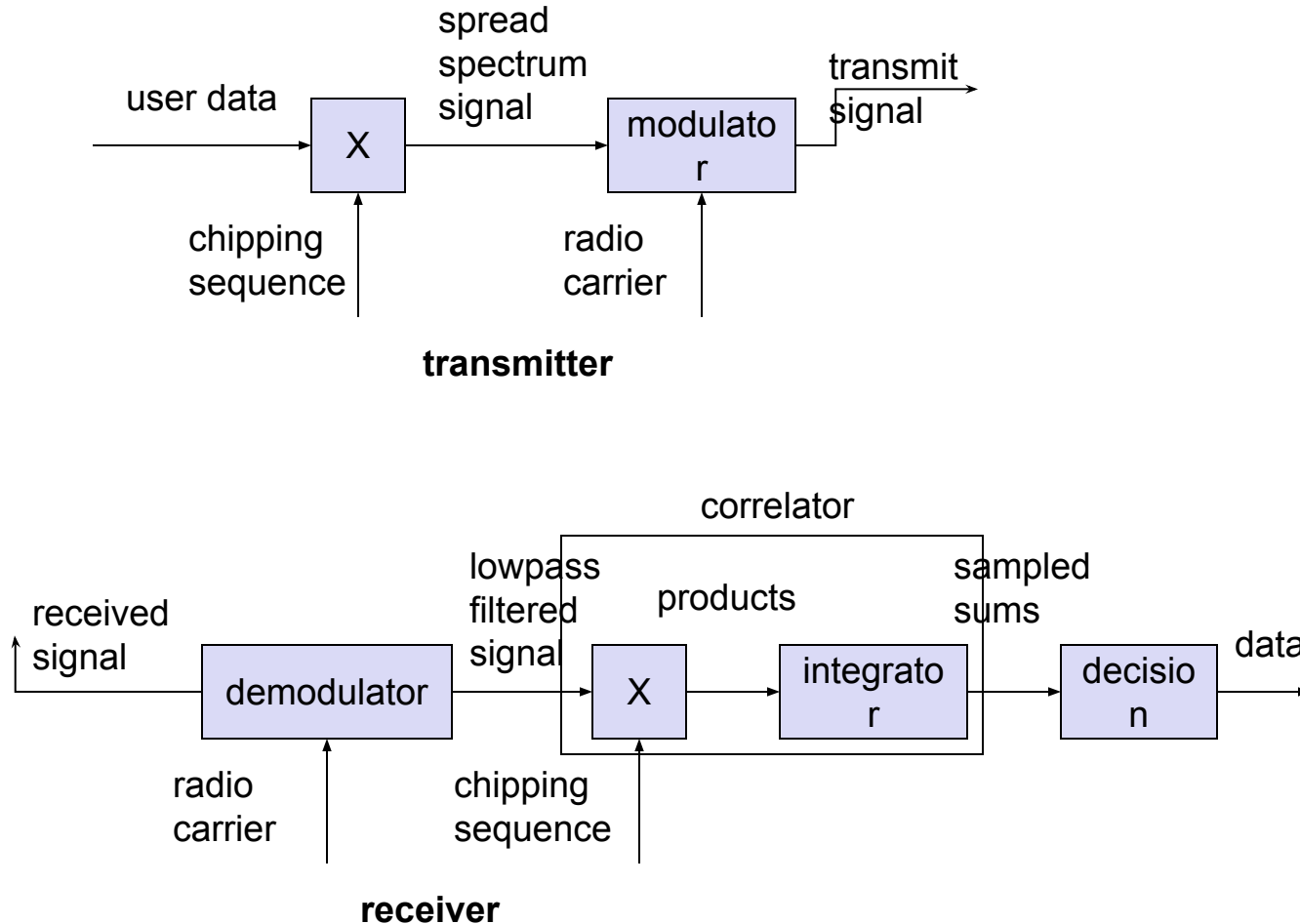


t_b : bit period
 t_c : chip period





DSSS (Direct Sequence Spread Spectrum) II





FHSS (Frequency Hopping Spread Spectrum) I

Discrete changes of carrier frequency

- ❑ sequence of frequency changes determined via pseudo random number sequence

Two versions

- ❑ Fast Hopping:
several frequencies per user bit
- ❑ Slow Hopping:
several user bits per frequency

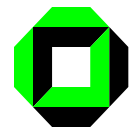
Advantages

- ❑ frequency selective fading and interference limited to short period
- ❑ simple implementation
- ❑ uses only small portion of spectrum at any time

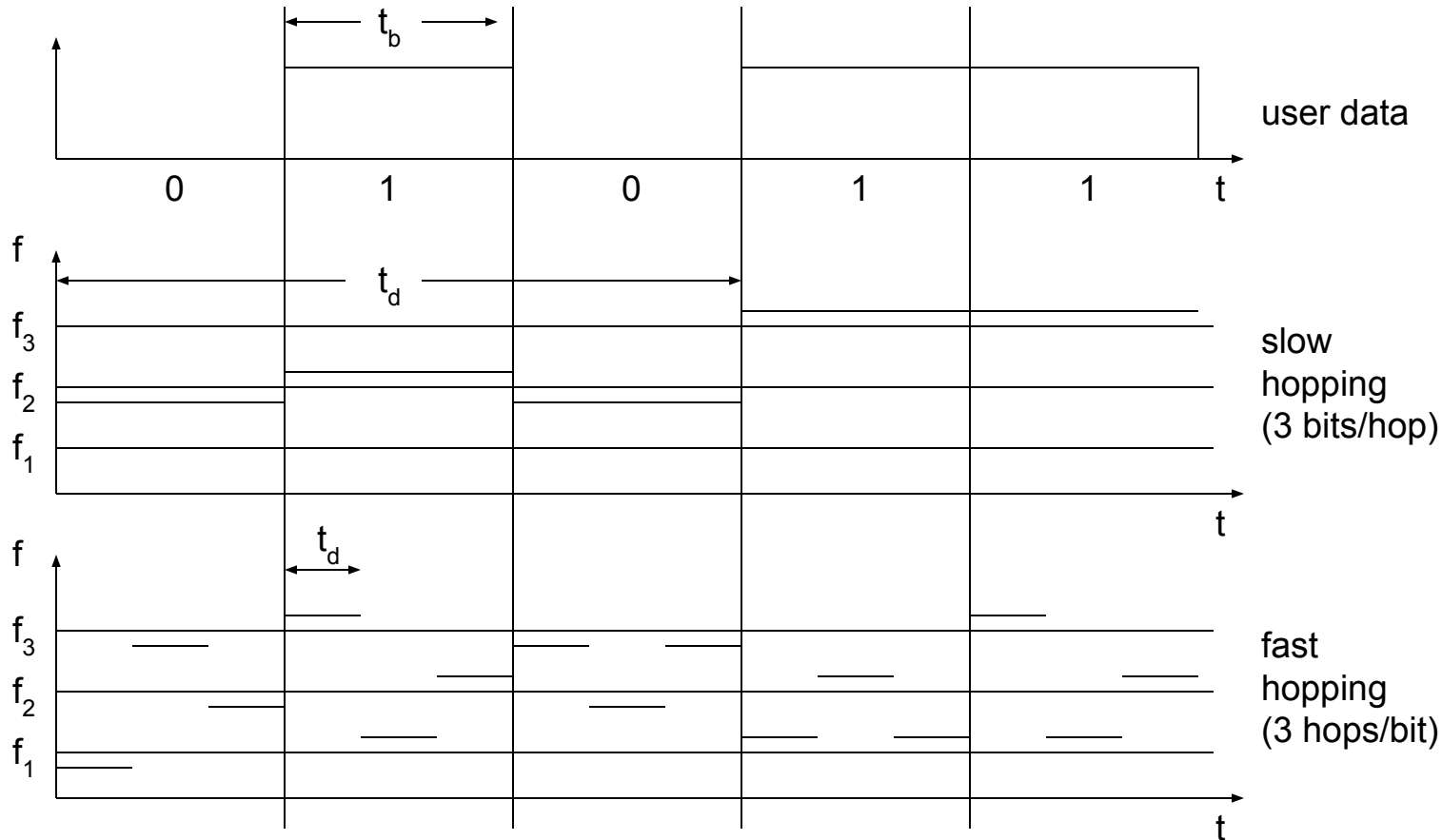
Disadvantages

- ❑ not as robust as DSSS
- ❑ simpler to detect





FHSS (Frequency Hopping Spread Spectrum) II

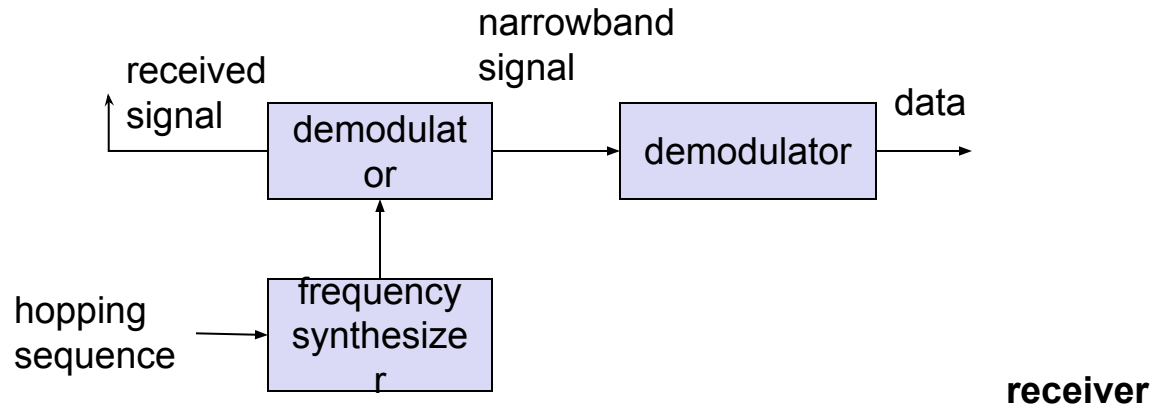
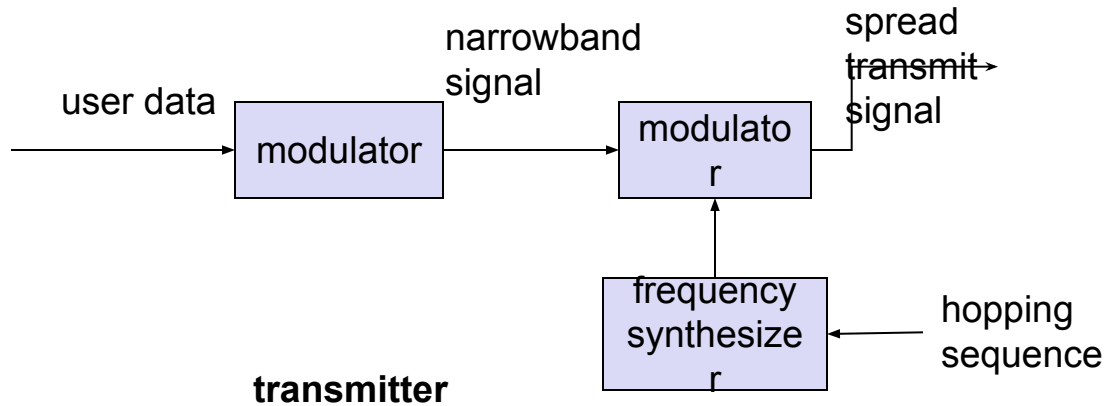


t_b : bit period t_d : dwell time





FHSS (Frequency Hopping Spread Spectrum) III





Cell structure

Implements space division multiplex: base station covers a certain transmission area (cell)

Mobile stations communicate only via the base station

Advantages of cell structures:

- ☐ higher capacity, higher number of users
- ☐ less transmission power needed
- ☐ more robust, decentralized
- ☐ base station deals with interference, transmission area etc. locally

Problems:

- ☐ fixed network needed for the base stations
- ☐ handover (changing from one cell to another) necessary
- ☐ interference with other cells

Cell sizes from some 100 m in cities to, e.g., 35 km on the country side (GSM) - even less for higher frequencies

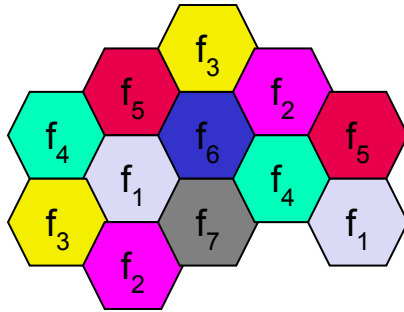




Frequency planning I

Frequency reuse only with a certain distance between the base stations

Standard model using 7 frequencies:



Fixed frequency assignment:

- ❑ certain frequencies are assigned to a certain cell
- ❑ problem: different traffic load in different cells

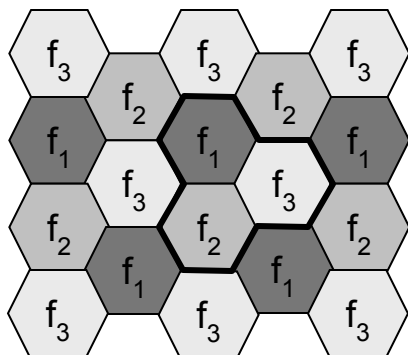
Dynamic frequency assignment:

- ❑ base station chooses frequencies depending on the frequencies already used in neighbor cells
- ❑ more capacity in cells with more traffic
- ❑ assignment can also be based on interference measurements



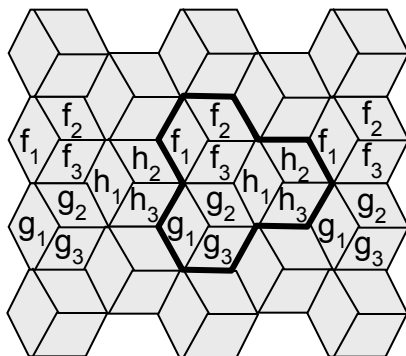
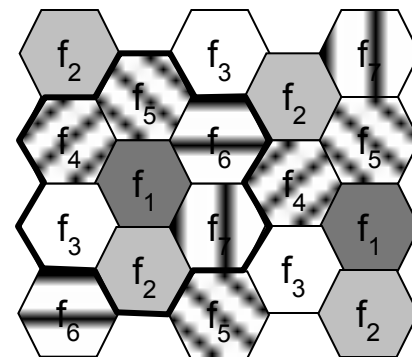


Frequency planning II



3 cell cluster

7 cell cluster



3 cell cluster
with 3 sector antennas

